

Bhoomika Lohana

Machine Learning Researcher | Data Science

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Summary:

I am an ML/AI Engineer and Researcher focused on building robust, evaluation-driven AI systems for real-world problems. My work involves data analysis, metric design, retrieval-augmented generation, and contextual engineering with large language models. I care about understanding what improves performance and why, so the systems I build work beyond controlled settings.

I approach problems from first principles. With a strong foundation in mathematics and statistics, I build AI systems end-to-end, starting from baseline performance, then iterating through meaningful evaluation and targeted optimization. My work includes LLM-in-the-loop systems, agentic pipelines with memory and self-evaluation, and multimodal retrieval frameworks, supported by rigorous error analysis.

Alongside engineering, I am actively involved in research on LLM reasoning, cultural bias, and creativity evaluation. I maintain a technical blog where I write about ML algorithms, mathematical intuition, and system design. I value clear thinking, structured execution, and building systems that hold up under real-world complexity.

Professional Experience

ML Researcher - 55mV ResearchLab (Remote – Australia)

April 2025 - Present

- Conducted controlled experiments to study how large language models perform mathematical reasoning across different cultural contexts, using culturally adapted variants of benchmark datasets.
- Designing reward signals and RL environments (SFT, GRPO) for the lab's first foundational model focused on creativity in advertising.
- Built embedding fusion strategies combining CLIP and SentenceTransformers to improve semantic retrieval in multimodal settings.
- Contributed to the development of a multimodal search system, now filed as a patent with IP Australia.
- Worked on retrieval-augmented generation pipelines, focusing on context design, error analysis, and real-world reliability of LLM-based systems.

Data Analyst- OULA

Jan - April 2025

- Analyzed market research data from 10+ product surveys to uncover consumer behavior patterns, supporting data-driven decisions across 3 business campaigns.
- Delivered actionable insights through Python reports and Power BI dashboards, influencing marketing strategies that improved engagement by 15%.

Research & Data Analyst Intern - Pakistan Hindu Council

June - August 2024

- Analyzed data from over 8,000 job fair participants to identify gender and regional trends, improving outreach and planning for future events.
- Assessed 120+ CSR activities and compiled impact reports to support minority-focused policy recommendations and national engagement efforts.

Exploratory Data Analyst Intern - Codexcue (Remote)

June– July 2024

- Analyzed e-commerce datasets from 5+ client projects using Python and Excel to identify sales trends, customer behavior, and performance metrics.
- Built interactive Power BI dashboards and applied clustering and regression techniques to support operational and marketing improvements.

Data Analyst Intern - InnoByte Services (Remote)

May – June 2024

- Analyzed and interpreted sales data to uncover trends, seasonality patterns, and key performance metrics, supporting data-driven strategic decisions.
- Applied data cleaning, transformation, and visualization techniques in Python (Pandas, Matplotlib) and Power BI to produce clear, accurate sales reports and actionable insights.

Publication

[Lost in Cultural Translation: Do LLMs Struggle with Math Across Cultural Contexts?](#)

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Co-authored a research project investigating how cultural context affects LLMs' mathematical reasoning, with equal contribution. I led the creation of six culturally localized versions of the GSM8K dataset by replacing entities (such as names, food, and places) with region-specific references while preserving mathematical logic. The dataset was released on Hugging Face (500+ downloads) to support reproducible evaluation of cultural bias in LLMs. Our findings show that LLMs, particularly smaller models, perform worse on culturally adapted problems, while culturally familiar models can outperform larger ones. This work highlights the importance of diverse training data in improving model robustness.

Dataset: [Hugging face](#)

[Galton's Law of Mediocrity: Why Large Language Models Regress to the Mean and Fail at Creativity in Advertising](#)

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Co-authored a research paper examining why large language models tend to regress toward average outputs and struggle with genuine creativity in advertising tasks. We frame this behavior as a form of regression to the mean and study it through controlled experiments on creative ad-generation settings. By analyzing model outputs and failure patterns, we show how current architectures favor safe, conventional ideas over truly novel ones. This work emphasizes the need for better evaluation frameworks and training strategies that can support more expressive, context-aware, and creatively diverse generative systems.

[Australian Patient: Document Search Engine](#)

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Designed a document search engine tailored for complex documents containing semantically linked components, such as advertising idea boards that include a headline, description, insights, and visuals. Traditional approaches store high-dimensional embeddings for each component and perform retrieval in that space, which is computationally expensive.

We proposed a more efficient solution. Instead of storing raw embeddings, we fuse all component embeddings into a single representation and transform it into a hashed value. Both documents and user queries are converted into hashed vectors, enabling fast retrieval via hash matching rather than high-dimensional similarity search. This method supports multimodal inputs such as image and text queries and consistently retrieves semantically closest matches. Experiments showed this approach is significantly faster and more accurate than traditional embedding-based retrieval systems.

Skills

- **Languages & Frameworks:** Python, SQL, PyTorch, RAG frameworks
- **ML & AI:** LLM fine-tuning, transformers Neural Networks, Econometrics (OLS, ARDL, ARIMA, VAR), Time series analysis, embedding techniques
- **Tools & Platforms:** EViews, STATA, Power BI, Excel, vector databases, multimodal learning frameworks
- **Other:** Statistical research design, hypothesis testing, data visualization, mathematical Modeling, Macroeconomic Data Analysis

Education

Bachelor of Science: Economics 2020-2024 | CGPA: 3.83

Lasbela University of Agriculture, Water, and Marine Sciences, Uthal - Balochistan, Pakistan.

Bachelor's Thesis – Empirical Time-Series Analysis

Impact of COVID-19 on the Pakistan Stock Market

- Used ARDL and OLS models to study how COVID-19 cases and key economic factors affected the KSE stock index in both the short and long run.
- Performed stationarity tests (ADF, KPSS, Phillips–Perron) and applied first-difference methods to ensure reliable time-series analysis.
- Included macroeconomic variables such as exchange rate, KIBOR, oil prices, and Economic Policy Uncertainty index to interpret economic and policy implications using real market data.

Certification

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| • Supervised Machine Learning: Regression and Classification | <u>Show Credentials</u> |
| • AI for Everyone by DeepLearning.ai | <u>Show Credentials</u> |
| • Data Analytics by IBM | <u>Show Credentials</u> |
| • EXCEL Specialization through Coursera. | <u>Show Credentials</u> |
| • Project Management | <u>Show Credentials</u> |
| • Business English: Networking through Coursera. | <u>Show Credentials</u> |